

Nick (Seungheon) Oh

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EDUCATION

Apr 2022 — Sept 2023	MSc Data Science and Artificial Intelligence, University of London Distinction	London
Sept 2016 — Jun 2020	BSc Government and Economics, London School of Economics and Political Science (LSE) Upper Second Class Honours [Cumulative GPA: 3.5-7/4.0]	London

PROFESSIONAL EXPERIENCE

Oct 2026 — Present	Researcher, London School of Economics and Political Science (LSE) Visiting researcher at CPNSS (Centre for Philosophy of Natural and Social Science), advised by Professor Fernand Gobet.	London
Jan 2023 — Present	Researcher, socius labs Founded socius labs , an independent research lab supported by the London School of Economics (LSE) and LSE Generate, bridging human sciences and artificial intelligence through two parallel research streams: applying insights from human sciences to enhance AI systems, and leveraging AI as a platform to run human theories at scale.	London
Mar 2025 — Sept 2026	Defence AI researcher, Ministry of National Defence Assigned to the S/C3 (Command, Control & Communications) Capability Requirements Branch, Training and Doctrine Command (TRADOC); on assignment from the Military Science and Technology Research Unit, Republic of Korea Army Headquarters [received TRADOC Commander's commendation for outstanding performance]	Republic of Korea
Nov 2021 — Oct 2022	Junior NLP engineer, Numen Capital Contributed to Knowsis research team extracting trading signals from non-traditional data sources (social media) through NLP and machine learning. Built transformer-based ESG (Environment, Social, Governance) tweet classifier achieving 97% accuracy, implemented rule-based sentiment analyser with custom financial dictionaries, and integrated these NLP-derived features into deep learning models for market movement forecasting. Developed end-to-end pipeline from raw text to actionable trading signals.	London

SELECTED PUBLICATIONS

Oh, N., & Gobet, F. (2026). Small Foundation Models of Human Cognition and Behaviour. *The 3rd Conference on Language Modeling (COLM'26)* [[Paper](#)]

Gobet, F., Bennett, D., & **Oh, N.** (2026). Discrimination Nets in Cognitive Science. *Thirteenth Annual Conference on Advances in Cognitive Systems (CogSys'26; Oral)*

Oh, N., & Gobet, F. (2025). Monitor-Generate-Verify (MGV): Formalising Metacognitive Theory for Language Model Reasoning. *Workshop on Foundations of Reasoning in Language Models @ the 39th Annual Conference on Neural Information Processing Systems (FoRLM Workshop, NeurIPS'25)* [[Paper](#)]

CONFERENCE
PROCEEDINGS

- Oh, N., & Gobet, F.** (2026). Small Foundation Models of Human Cognition and Behaviour. *The 3rd Conference on Language Modeling (COLM'26)* [[Paper](#)] [[Poster](#)]
- Gobet, F., Bennett, D., & **Oh, N.** (2026). Discrimination Nets in Cognitive Science. *Thirteenth Annual Conference on Advances in Cognitive Systems (CogSys'26; Oral)*
- Oh, N., Vrakas, G. D., Brooke, S. J., Morinière, S., & Duke, T.** (2025). PETLP: A Privacy-by-Design Pipeline for Social Media Data in AI Research. *Proceedings of the 8th AAAI/ACM Conference on AI, Ethics, and Society (AIES'25)* [[Paper](#)] [[Poster](#)]
- Oh, N.** (2024). Sentibank: A Unified Resource of Sentiment Lexicons and Dictionaries. *Proceedings of the 18th International AAAI Conference on Web and Social Media (ICWSM'24)* [[Paper](#)] [[Poster](#)]
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WORKSHOPS

- Oh, N., & Jin, H.** (2026). Reliability, Faithfulness, and the Limits of Post-hoc Explanations of Opaque Scientific Models. *Philosophy of Machine Learning Workshop @ the 43rd International Conference on Machine Learning (PhilML Workshop, ICML'26)* [[Paper](#)] [[Poster](#)]
- Oh, N., & Gobet, F.** (2025). Monitor-Generate-Verify (MGV): Formalising Metacognitive Theory for Language Model Reasoning. *Workshop on Foundations of Reasoning in Language Models @ the 39th Annual Conference on Neural Information Processing Systems (FoRLM Workshop, NeurIPS'25)* [[Paper](#)] [[Poster](#)]
- Oh, N.** (2025). Before you <think>, monitor: Implementing Flavell's metacognitive framework in LLMs. *Workshop on the Application of LLM Explainability to Reasoning and Planning @ Conference on Language Modeling (XLLM-Reason-Plan Workshop, COLM'25)* [[Paper](#)] [[Poster](#)]
- Oh, N., & Gobet, F.** (2024). System 1.5: Designing Metacognition in Artificial Intelligence. *System 2 Reasoning at Scale Workshop @ the 38th Annual Conference on Neural Information Processing Systems (Sys2-Reasoning Workshop, NeurIPS'24; Spotlight)* [[Paper](#)] [[Poster](#)] [[Spotlight Presentation \(Video\)](#)]
- Oh, N.** (2024). In Defence of Post-hoc Explainability. *Interpretable AI: Past, Present and Future Workshop @ the 38th Annual Conference on Neural Information Processing Systems (IAI Workshop, NeurIPS'24)* [[Paper](#)] [[Poster](#)]
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WORKING PAPERS

- Thinking Without Words: The Representational Geometry of Latent Reasoning Trajectories, with Cho, Y., Lee, M., Joo, S. & Lee, A.
TL;DR: Tests whether purely latent chains of thought can sustain long-horizon reasoning by characterising the representational geometry of their trajectories.
- The Ideological Subspace of Large Language Models: Angular Steering of Political Representations, with Cho, Y. & Gobet, F.
TL;DR: Locates an ideological subspace in LLMs, characterises its structure, and uses angular steering to shift models' political positions.
- Do Models of the Mind Converge? Representational Analysis of LLMs Fine-tuned on Human Behaviour, with Gobet, F.
TL;DR: Traces how representations in small foundation models of human cognition evolve with LoRA rank and data scale, and whether they converge across model sizes and families.
- Mapping the Architecture–Task Space of Neural Network Models of Human Behaviour
TL;DR: A survey charting which neural network architectures have been trained on which behavioural tasks.
- High-Intensity Ambivalence: Beyond Acceptance and Resistance to Generative AI in Occupational Communities (in review), with Humphrey, J. & Ross, D.
TL;DR: Argues that occupational communities' responses to generative AI are better understood as high-intensity ambivalence than as acceptance or resistance.

TEACHING & TUTORIALS	<i>HAI101: Human x Artificial Intelligence</i> — Lecture series @ LSE CPNSS (2026), organiser and speaker [Website] <i>Introduction to Transformers and LLMs</i> — Lecture note (2026), with Feng, S. [PDF] <i>Scraping Reddit the Right Way: A Guide to Legal and Ethical Data Collection with RedditHarbor</i> — Tutorial @ ICWSM'24, co-organised with the Open Data Institute (ODI)	
EXHIBITIONS	Oh, N., & Park, A. (2026). artefact(s): LeNet-1. <i>Art Gallery @ the IEEE/CVF Conference on Computer Vision and Pattern Recognition, Denver (CVPR'26; IEEE TCPAMI Art Award; Spotlight)</i> [Gallery] [Video]	
RELEVANT TRAINING & SKILLS	LSE MA314: Algorithms and Programming (First Class) LSE GV324: Applied Quantitative Methods for Political Science (First Class) LSE GV249: Research Design in Political Science (Upper Second Class)	LSE Methodology & Quantitative Training: Causal inference and natural experiments in the social sciences (regression, matching, panel data, diff-in-diff, instrumental variables, regression discontinuity) Python: PyTorch, spaCy, NLTK, transformers, trl, unsloth, vllm, peft, PRAW, OpenAPI